A representation of this quiver consists of two vector spaces V, W and an operator $A : V \to W$:

$$\underset{V}{\bullet} \xrightarrow{A} \underset{W}{\bullet}.$$

To decompose this representation, we first let V' be a complement to the kernel of A in V and let W' be a complement to the image of A in W. Then we can decompose the representation as follows:

$$\underset{V}{\bullet} \xrightarrow{A} \underset{W}{\bullet} \;=\; \underset{\ker A}{\bullet} \xrightarrow{0} \underset{0}{\bullet} \;\oplus\; \underset{V'}{\bullet} \xrightarrow{\overset{A}{\sim}} \underset{\operatorname{Im} A}{\bullet} \;\oplus\; \underset{0}{\bullet} \xrightarrow{0} \underset{W'}{\bullet}.$$

The first summand is a multiple of the object $1 \longrightarrow 0$, the second a multiple of $1 \longrightarrow 1$, and the third of $0 \longrightarrow 1$. We see that the quiver A_2 has three indecomposable representations, namely

$$1 \longrightarrow 0\,, \quad 1 \longrightarrow 1\,, \quad \text{and} \quad 0 \longrightarrow 1\,.$$

Note that this statement is just the Gauss elimination theorem for matrices.

Example 6.2.4 (A_3)**.** The quiver A_3 consists of three vertices and two connections between them. So we have to choose between two possible orientations:

$$\bullet \longrightarrow \bullet \longrightarrow \bullet \quad \text{or} \quad \bullet \longrightarrow \bullet \longleftarrow \bullet\,.$$

(1) We first look at the orientation

$$\bullet \longrightarrow \bullet \longrightarrow \bullet\,.$$

Then a representation of this quiver looks like

$$\underset{V}{\bullet} \xrightarrow{A} \underset{W}{\bullet} \xrightarrow{B} \underset{Y}{\bullet}\,.$$

As in Example 6.2.3 we first split away

$$\underset{\ker A}{\bullet} \xrightarrow{0} \underset{0}{\bullet} \xrightarrow{0} \underset{0}{\bullet}\,.$$

This object is a multiple of $1 \longrightarrow 0 \longrightarrow 0$. Next, let Y' be a complement of $\operatorname{Im} B$ in Y. Then we can also split away

$$\underset{0}{\bullet} \xrightarrow{0} \underset{0}{\bullet} \xrightarrow{0} \underset{Y'}{\bullet}$$